

Detecting Volatility Regimes in Crypto Markets Using Realized Volatility Structure and Normalized Momentum

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Disclosure:

The author currently works in broker technology solutions with a financial technology company in India. The views and analysis presented in this paper are solely those of the author and do not represent the views of any employer or affiliated organization.

Abstract

This paper proposes a simple, strategy-agnostic framework for detecting market regimes in cryptocurrency markets using realized volatility structure and normalized directional momentum. Instead of forecasting returns or optimizing trading strategies, the framework classifies each moment in time into one of three structural regimes—Expansion, Neutral, and Contraction—based on the relationship between short-term and long-term realized volatility and the strength of directional movement relative to recent history. Using high-frequency BTC data from 2024 and 2025 YTD, the study demonstrates that these regimes exhibit stable distributions and materially different volatility and tail-risk characteristics. The results suggest that regime detection based on realized volatility structure can serve as a robust control layer for downstream trading and risk-management systems.

1. Introduction

Cryptocurrency markets are often approached through indicator-driven or strategy-centric frameworks that implicitly assume homogeneous market behavior across time. In practice, however, the performance and risk profile of any strategy is strongly dependent on the prevailing volatility environment. Strategies that perform well during periods of volatility expansion often degrade or fail during contractionary or balanced market states.

This motivates the need for an explicit market-state detection layer that operates independently of trading logic. Rather than embedding regime assumptions implicitly within strategies, this paper proposes a regime-first framework designed to identify structural market states using realized volatility dynamics and normalized directional pressure. The objective is not to generate trading signals, but to classify market conditions in a manner that is stable, interpretable, and suitable as a control layer for downstream systems.

2. Conceptual Framework

The framework rests on two core observations.

First, market behavior is better characterized by **changes in volatility structure** than by absolute volatility levels. Volatility expansion and contraction reflect shifts in market participation, liquidity, and risk appetite that are not captured by static thresholds.

Second, directional price movement is only meaningful when evaluated **relative to the market's own recent behavior**. Raw momentum measures are scale-dependent and fail to distinguish between statistically significant moves and routine noise.

To address these issues, the framework combines:

- **Realized Volatility Structure:** the relationship between short-term and long-term realized volatility, capturing whether volatility is expanding, contracting, or stable relative to its recent baseline.
- **Normalized Directional Momentum:** recent directional price movement expressed as a standardized measure relative to historical variation.

Together, these dimensions define three regimes:

- **Expansion:** accelerating volatility accompanied by directional commitment
- **Neutral:** balanced volatility with weak or statistically insignificant directional dominance
- **Contraction:** compressing volatility and suppressed directional movement

The regime classification is applied continuously at a fixed time resolution and is independent of any specific trading strategy.

3. Data and Methodology

The analysis uses minute-level BTC spot and futures data covering the full year 2024 and 2025 year-to-date, totaling approximately 471,000 observations.

Realized volatility is computed from futures returns using rolling windows to estimate short-term and long-term volatility. Directional momentum is measured over a short lookback window and normalized using z-score transformations over a longer historical window to ensure scale-free comparability across time.

Regime classification is performed using fixed thresholds applied consistently across the dataset. No parameter optimization or outcome-based tuning is performed, and the regime logic is held constant across all tests.

4. Empirical Results

Across the full sample period, the regime distribution is stable:

- Neutral regimes account for approximately 62% of observations
- Expansion regimes account for approximately 20%
- Contraction regimes account for approximately 18%

The regimes exhibit clearly differentiated structural characteristics. Expansion regimes are associated with higher short-term realized volatility and significantly heavier tail behavior, as measured by 95th and 99th percentile moves. Contraction regimes exhibit compressed volatility and reduced tail risk, while Neutral regimes sit structurally between the two.

These patterns persist across months and across varying market conditions, indicating that the regime classification captures persistent structural properties rather than transient noise.

5. Implications

The results suggest that regime detection based on realized volatility structure provides a robust and interpretable market-state layer. By explicitly separating regime identification from trading logic, the framework enables downstream strategies to be conditioned on market state rather than implicitly optimized across heterogeneous environments.

This regime-first approach can improve risk control, reduce strategy fragility, and support cleaner system architectures across directional, carry, and volatility-focused trading applications.

Importantly, the framework is not presented as a trading strategy, but as a control mechanism designed to govern strategy behavior and risk exposure.

6. Conclusion

This paper demonstrates that a small set of interpretable measures—realized volatility structure and normalized directional momentum—are sufficient to identify persistent market regimes in cryptocurrency markets. By avoiding strategy-specific assumptions and focusing on structural behavior, the proposed framework provides a scalable foundation for institutional-grade trading and risk-management systems.

Future work may explore extensions to other asset classes, additional conditioning variables, or the interaction between regime states and specific strategy families.